

Speech Digit Recognition Project Roadmap

Project Goal: Build a classifier that can recognize spoken digits (0-9) from audio recordings

Completion Target: May 3rd, 2025

Week 1 (April 5-11): Setup & Data Preparation

- **Day 1-2:** Environment setup
 - Install required libraries (librosa, numpy, scikit-learn, tensorflow/pytorch)
 - Download Free Spoken Digit Dataset from GitHub
 - Set up project structure and version control
- **Day 3-4:** Data exploration
 - Analyze audio files (duration, sampling rate, speakers)
 - Visualize waveforms and spectrograms
 - Split data into training/validation/test sets
- **Day 5-7:** Feature extraction pipeline
 - Implement MFCC extraction
 - Extract other potentially useful features (zero-crossing rate, spectral centroid)
 - Normalize and save features for faster access during training

Deliverables Week 1:

-  Initialized GitHub repository with README.md
-  Dataset exploration report (statistics, visualizations)
-  Feature extraction module (Python script)
-  Preprocessed dataset (saved features in appropriate format)

Week 2 (April 12-18): Model Building & Initial Training

- **Day 1-2:** Baseline model implementation
 - Build simple SVM classifier with basic MFCC features
 - Establish performance baseline
- **Day 3-4:** Feature engineering
 - Experiment with feature selection/transformation
 - Compare different audio preprocessing techniques

- **Day 5-7:** Basic model improvements
 - Try different classifiers (Random Forest, KNN)
 - Implement cross-validation
 - Document results and identify promising approaches

Deliverables Week 2:

-  Baseline model implementation (Python script)
-  Model evaluation report (baseline performance metrics)
-  Feature engineering analysis document
-  Comparison of at least 3 different traditional ML models

Week 3 (April 19-25): Advanced Models & Optimization

- **Day 1-3:** Neural network implementation
 - Design simple CNN or RNN architecture for audio classification
 - Train initial deep learning models
- **Day 4-5:** Hyperparameter tuning
 - Optimize model parameters for best performance
 - Experiment with different architectures if using neural networks
- **Day 6-7:** Model evaluation
 - Compare all implemented models
 - Analyze confusion matrix and error patterns
 - Select best performing model(s)

Deliverables Week 3:

-  Neural network model implementation (Python script)
-  Hyperparameter tuning report
-  Comprehensive model comparison document
-  Best model checkpoint/saved model file

Week 4 (April 26-May 3): Refinement & Final Deliverables

- **Day 1-2:** Final model improvements
 - Ensemble methods if beneficial
 - Error analysis and targeted improvements

- **Day 3-4:** Inference pipeline
 - Build real-time prediction functionality
 - Optimize for speed if needed
- **Day 5-7:** Documentation & finalization
 - Write detailed documentation of approach and results
 - Create visualization of model performance
 - Package project for easy demonstration
 - Final testing and bug fixes

Deliverables Week 4:

-  Final optimized model
-  Inference pipeline for real-time digit recognition
-  Performance visualization dashboard
-  Complete project documentation including:
 - Methodology report
 - Results analysis
 - User guide for model deployment/usage
-  Final presentation slides

Technical Focus Areas

- 1. Feature extraction techniques:**
 - MFCCs (Mel-frequency cepstral coefficients)
 - Spectral features (centroid, bandwidth, contrast)
 - Temporal features (zero-crossing rate, RMS energy)
- 2. Classification approaches to explore:**
 - Traditional ML: SVM, Random Forest, KNN
 - Neural networks: Small CNN, RNN/LSTM, or GRU architectures
- 3. Evaluation metrics:**
 - Accuracy, precision, recall, F1-score
 - Confusion matrix analysis
 - Cross-validation performance

Final Project Deliverables (Due May 3rd)

1. **Code:**

- Well-documented Python modules for each project component
- Requirements.txt file for environment setup
- Notebook demonstrations of key functionality

2. **Models:**

- Trained model files for best performing classifiers
- Model card describing performance characteristics

3. **Documentation:**

- Technical report (PDF) explaining approach, methods, and results
- README with setup and usage instructions
- Performance analysis and visualization

4. **Demo:**

- Simple web or command-line interface for live digit classification
- Example audio files for testing